

Nahid Hasan Sagar

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PROFESSIONAL SUMMARY

M.S. candidate in Statistical Science at Indiana University Bloomington with strong quantitative training in statistical modeling, computational analysis, and evidence-based decision support. Experienced in building and validating high-stakes analytical workflows, synthesizing complex outputs into clear recommendations, and collaborating across teams in fast-paced environments.

EDUCATION

Indiana University, College of Arts and Sciences

Master of Science in Statistical Science

Bloomington, IN

Expected May 2027

Thesis: Time-Series Analysis Using Bayesian Additive Regression Trees (BART)

Graduate Coursework: Introduction to Statistical Computing; Applied Statistical Computing; Fundamentals of Statistics I–II; Applied Linear Models I–II; Statistical Consulting; Research in Statistics

University of Dhaka

Bachelor of Science in Physics

Dhaka, Bangladesh

August 2024

RELEVANT EXPERIENCE

Research Assistant, StatMIND Lab

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Perform in-depth quantitative analysis on high-dimensional datasets, translating modeling outputs into clear, decision-oriented findings for research stakeholders.
- Implement and validate Bayesian computational routines in C++ using Rcpp and RcppArmadillo, ensuring numerical accuracy through unit testing and reproducible checks.
- Built structured benchmarking and documentation workflows to evaluate performance tradeoffs across implementations, improving analytical transparency and speed.
- Delivered a 4 to 5× speedup in core computations by moving numerically intensive work into compiled C++ code while preserving model reliability.

Teaching Assistant, Department of Statistics

Indiana University Bloomington

2025 – Present

Bloomington, IN

- Communicate complex quantitative methods to diverse audiences through discussion sections, office hours, and targeted problem-solving support.
- Evaluate approximately 200 assignments per cycle with strong attention to detail, consistency, and quality control under tight timelines.
- Present structured feedback on statistical reasoning and interpretation, strengthening data-driven communication and analytical discipline.

Senior Mathematics Instructor

Udvash Academic & Admission Care

2016 – 2025

Dhaka, Bangladesh

- Taught and mentored more than 100,000 students through classroom instruction and national online learning platforms.
- Managed multiple concurrent priorities, including curriculum planning, instruction delivery, and progress tracking in high-volume settings.
- Developed concise explanatory materials and presentations to communicate technical content with clarity and impact.

SELECTED PROJECTS

M.S. Thesis: Time-Series Analysis Using BART

R, Bayesian Modeling

- Conduct research on Bayesian Additive Regression Trees for time-series analysis, evaluating predictive performance and model behavior under a rigorous statistical framework.

Ordinary Least Squares from Scratch

Python, NumPy, Pandas, Matplotlib

- Developed an implementation of ordinary least squares regression from first principles, emphasizing matrix-based estimation, numerical validation, and clear model interpretation.

TECHNICAL SKILLS

Quantitative Methods: Statistical inference, Bayesian modeling, time-series analysis, regression, model validation, predictive evaluation, experimental design, and computational statistics

Programming: R, Python, C++, MATLAB, Bash, SQL

Tools and Platforms: Git, GitHub, LaTeX, RStudio, VS Code, Jupyter, NumPy, Pandas, Matplotlib

Languages: Bengali (native), English (fluent), Hindi (conversational), Urdu (conversational)

LEADERSHIP & ACTIVITIES

- Attended the 2026 Velocity Conference at UC Berkeley as an Indiana University representative for the science community.
- Co-organized the Australian Mathematics Competition in Bangladesh in 2014.